

GRAVITATIONAL CLUSTERING

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(Received 10 February 1976; in revised form 7 March 1977)

Abstract—This paper introduces and describes an algorithm or technique, called gravitational clustering, for performing cluster analysis on Euclidean data. The paper describes the physical gravitational model, an abstract generalized model, and several specific gravitational models. It illustrates clustering by one of these models using several sample data sets, and compares the results with those obtained using two other well-known nongravitational clustering methods. The paper also illustrates four graphical techniques to aid in the analysis of a clustering.

Cluster analysis Gravitational clustering Classification Contour plot Euclidean data

INTRODUCTION

Cluster analysis may be defined intuitively as the partitioning of a set of elements into subsets, called clusters, so as to maximize similarity within clusters and minimize similarity between clusters. Many heuristic methods have been set forth to carry out this general notion of clustering, each having certain apparent advantages and disadvantages, and each appearing to conform more or less to this general notion. This paper is concerned with another such method, inspired by the motion of particles in space due to their gravitational attraction for each other, and called gravitational clustering.

There is a phenomenon in physics in which a finite system of particles in space, each with a specified initial location, a zero velocity, a given mass, and a negligible volume, converges to the centroid of the system due to the gravitational attraction between the masses. As the particles travel toward the centroid they join through inelastic collisions with others closest to them, forming what may be considered to be new, conglomerate, particles. This process results in a system of conglomerate particles which are increasing in size, decreasing in number, and converging toward the centroid. Eventually there will be three conglomerate particles remaining, then only two, and finally just one, with mass equaling the sum of the masses of the original particles, and located at the centroid.

This natural phenomenon gives rise to a general method of clustering which shall be called “gravitational clustering”. The implementation of the clustering algorithm consists of simulating the movement of the particles from their original positions to the final position, and analyzing this movement to determine the clustering results. Throughout the life of the system, each remaining particle corresponds to a cluster. Initially there will be as many clusters as there are original particles, with each cluster being comprised of a single particle. Whenever two particles join to form a new particle, the cluster corresponding to

the new particle will be the union of the clusters corresponding to the joining particles. The last two clusters remaining, whose union is the entire set of original particles, define the “optimal” 2-level clustering. The last three clusters remaining define the “optimal” 3-level clustering, and so on for 4, 5, etc.

This method is clearly an agglomerative, hierarchical algorithm such as described in Ward.⁽⁹⁾ There is, however, no objective function which is minimized or maximized at each stage. Instead, the joining of particles is determined by the continuous motion of all particles in the system according to the gravitational forces. Coleman⁽²⁾ discusses a method of clustering which also uses a system of forces, although it is otherwise unlike this method. The attributes of time and mass are not a part of the model (although mass is mentioned briefly), and particles do not converge together but rather converge to “equilibrium” positions. Coleman’s model is based on attributes called “psychological distances” and employs repulsive as well as attractive forces.

The element of time in the gravitational model is very important, adding a new dimension to the cluster analysis. As was mentioned previously, the sequence in which the particles come together defines the various clusterings, but the times at which such joinings occur, and the intervals between the joinings, give additional insight into the relative strength of each cluster and clustering.

The general principle is that the longer the system exists with a certain number of particles (clusters), the greater is the strength of the clustering at that level. Similarly, the greater the length of time between the creation of a cluster and its union with another cluster, the greater the strength of the former. A normalizing factor is the total time required for the system to be reduced to one cluster.

As an example, suppose that a system of 100 particles combines very rapidly at first so that at time 15 sec (or other unit) there are 3 clusters remaining. Now suppose that these clusters are relatively far apart, and that 2 of them do not join until time 65 sec.

Finally, suppose that the 2 clusters remaining join fairly quickly at time 73 sec. This model would indicate then that the clustering with the greatest strength is at the 3 level and that it is very strong, having accounted for $(65 - 15)/73 \cdot 100\% = 68\%$ of the total life of the system.

The comparison of the strengths of clusterings can be extended to comparisons between different data sets. For example, with one data set the optimal clustering may be at the 3 level and may occupy 68% of the life of the system. With another data set, the optimal clustering may be at the 2 level and may occupy only 45% of the system life. This would indicate that the first system clusters much better than the second.

It is important to note that while true gravitational motion has served as a model for this clustering method, the method has not been limited to the use of all the various physical laws involved. The purpose of the method is to do cluster analysis, and there are no theoretical or logical restrictions against modifying the method so as to more completely implement notions of clustering.

With this generalized view of the model, the movement of each particle during each time interval is given by a function g , which has been termed a *gravitational function*. The specification of a gravitational function thus determines a specific gravitational clustering model. Obviously, not all gravitational functions give rise to models which yield reasonable results from the point of view of cluster analysis, and the entire concept of gravitational clustering is dependent on whether or not there exist one or more gravitational functions which yield good clustering results. There do in fact exist such functions, one of which will be illustrated in some detail later.

The "simulation" aspect of the model arises because the movement of the particles is approximated over small discrete time intervals $[t, t + dt]$, according to the forces acting on the particles at the beginning of the intervals. In theory, the forces are a function of continuous time, and motion of the particles occurs continually rather than in discrete jumps. However, it is quite feasible to get very good approximations of continuous motion using this discrete simulation.

The procedure currently implemented determines the time increment dt at each step so as to permit the fastest moving particle to move a distance δ , which is a parameter of approximation. This parameter should be set small enough that the simulated motion is substantially the same as the theoretical motion would be. However, since a smaller value of δ will require a greater amount of computer processing time, δ should not be set unnecessarily small. Simple experimentation, with regard for the separation of the original data points, has been a satisfactory procedure for determining δ thus far. For large problems it may be necessary to compromise between speed and accuracy in setting δ .

Critical points in the process occur whenever two

particles come together. When this happens the forces acting on the two particles approach infinity in magnitude, as do the velocities and accelerations of the particles. This phenomenon, however, causes neither theoretical nor practical problems in the clustering model. Although these force, velocity, and acceleration vectors approach infinity in magnitude, their sums are finite since they are almost opposite in direction (the influence of other particles keeps them generally from being exactly opposite). Assuming they combine into a single particle on impact, the motion of this particle immediately after impact is finite, since the infinite components cancel out.

Treatment of this occurrence in the practical model is very straightforward. Since the particles coming together have velocities approaching infinity, the time for them to collide, starting from a short distance ϵ apart, approaches zero. The rest of the particles, of course, are virtually motionless during such a short period (except when there are multiple collisions). An approximate implementation of a collision can therefore be made by first recognizing when two particles are within ϵ distance of each other, and then replacing them by a single particle located at their centroid. Because of the influence of other particles, the collision would not in general occur exactly at the centroid, but the difference can be made negligible by making ϵ small. When there are multiple collisions occurring at the same time, they can be treated as a sequence of collisions at that time. The value of ϵ is currently specified as a parameter of approximation, usually $2 \cdot \delta$.

THE FORMAL MODEL AND SOME EXAMPLES

The formal model

The true gravitational model described briefly in the previous section is a type of clustering procedure, in that it starts with a set of points and winds up grouping them into various clusters "so as to maximize similarity within clusters and minimize similarity between clusters" in some sense. There is, however, no justification for assuming that the particular relations, formulas, and operating characteristics in that model would be appropriate for every or any situation in which clustering is desired. But—the general operation of the model, by which *original elements move continuously "toward" each other according to a specified "gravitational" relation, joining other elements along the way to form "clusters" which combine ultimately into one*, is a very useful model which shall be called a "*gravitational clustering*" procedure. The formal gravitational clustering procedure is as follows:

(a) A finite number n of elements or particles $p_1, \dots, p_n$ are originally specified by their locations or measurements $s_1, \dots, s_n$, and by their masses $m_1, \dots, m_n$. It is assumed that all measurements have been appropriately scaled so as to yield this set of n points

in Euclidean m -space. No scaling problems are considered here.

(b) Two parameters of approximation, δ and ϵ , must be specified. In each time interval the length dt of the interval will be determined so that the fastest moving particle will move the distance δ . When two particles come within ϵ distance of each other, they are joined into a single particle at their centroid. A good value for ϵ is $2 \cdot \delta$.

(c) The time t is initialized to 0. Steps d, e, and f are repeatedly executed until only one distinct particle remains.

(d) The movement of each remaining particle i during the interval $[t, t + dt]$ is computed according to the function $g(i, t, dt)$, which is a function of the attributes of particle i and all other remaining particles. The time increment dt is determined so that the longest distance moved will be δ (there are other possible ways for determining dt or specifying δ).

(e) The new position $s_i(t + dt)$ of each remaining particle i is computed as the sum $s_i(t + dt) \leftarrow s_i(t) + g(i, t, dt)$. The time t is incremented by dt , i.e. $t \leftarrow t + dt$.

(f) If any pair of particles i and j have moved to within ϵ distance of each other, then particle j is joined into particle i (assume $i < j$). Specifically, $s_i(t) \leftarrow (m_i(t) \cdot s_i(t) + m_j(t) \cdot s_j(t)) / (m_i(t) + m_j(t))$, $m_i(t) \leftarrow m_i(t) + m_j(t)$, and particle j is deleted.

The physical model reexamined

Now that the definition of a gravitational clustering model has been given, it will be shown that the physical model discussed earlier is in fact a gravitational model. To do this it is necessary only to give the function $g = g(i, t, dt)$, which gives the movement of particle i at time t . The following development does this (assume $N(r)$ is the set of particles remaining at time r):

$$g(i, t, dt) = v_i(t) dt + 1/2 a_i(t) dt^2 \quad (1)$$

$$v_i(t) = \int_0^t a_i(r) dr$$

$$a_i(r) = \sum_{j \in N(r), j \neq i} G \frac{m_i(r)m_j(r)}{m_i(r)} \frac{1}{|s_j(r) - s_i(r)|^2} \times \frac{s_j(r) - s_i(r)}{|s_j(r) - s_i(r)|}$$

$$g(i, t, dt) = \int_0^t \left[G \sum_{j \in N(r), j \neq i} \frac{m_i(r)m_j(r)}{m_i(r)} \frac{1}{|s_j(r) - s_i(r)|^2} \times \frac{s_j(r) - s_i(r)}{|s_j(r) - s_i(r)|} \right] dr dt + 1/2 G \sum_{j \in N(t), j \neq i} \frac{m_i(t)m_j(t)}{m_i(t)} \times \frac{1}{|s_j(t) - s_i(t)|^2} \frac{s_j(t) - s_i(t)}{|s_j(t) - s_i(t)|} dt^2.$$

The final equation above gives the gravitational function $g(i, t, dt)$, therefore the physical model is in fact a gravitational clustering model.

A modification to the physical model

As mentioned earlier, there has been no justification for assuming that the particular operating characteristics of the physical model are appropriate for any or every clustering problem. Thus, it is of interest to consider modifications to the model which might improve its properties or performance. One such modification imparts to the model a generally highly desirable property, which shall be called *Markovian*.

A gravitational model will be said to be a Markovian model if the function $g_{i,dt}(t) = g(i, t, dt)$, considered as a function of t , depends only on the present attributes (locations and masses) of the remaining particles and not on any past history. By way of clarification, let P denote a set of particles (the location and mass of each particle). For $t \geq 0$ let $P(t)$ denote the set at time t in a gravitational clustering of $\dot{P}$. Clearly, $P(0) = P$. Now suppose $u \geq 0$ and $P(u) = Q$ for some set of particles Q . Then for a Markovian gravitational model, $P(t) = Q(t - u)$ for every $t \geq u$.

The physical model is not Markovian because the gravitational function at a particular time t depends not only on the locations and masses of the particles at that time but also on the velocities which they have built up. A Markovian model can be obtained from the physical model by merely deleting the velocity term in the motion formula (1). The net effect on the formal model is that the gravitational function $g(i, t, dt)$ becomes

$$g(i, t, dt) = 1/2 G \sum_{j \in N(t), j \neq i} \frac{m_i(t)m_j(t)}{m_i(t)} \frac{s_j(t) - s_i(t)}{|s_j(t) - s_i(t)|^3} dt^2. \quad (2)$$

This new model is analogous to motion in a *viscous fluid*, in which particles cannot build up any momentum. Each particle moves directly toward its net attraction, without any orbiting or oscillation. As mentioned, the true physical model does not possess these qualities. While its particles may ultimately meet at the centroid, they might have oscillating and orbiting tendencies along the way. It is obvious that with the velocity component deleted, as in the modified model, each particle will necessarily move toward its net attraction since it has no influence otherwise.

The generalized Markovian model

A slight generalization of (2) gives a class of gravitational models which shall be called the generalized Markovian model. The gravitational function is

$$g(i, t, dt) = dt^2 \sum_{j \in N(t), j \neq i} \frac{m_i^p(t)m_j^q(t)}{m_i(t)} \frac{1}{|s_j(t) - s_i(t)|^2} \times \frac{s_j(t) - s_i(t)}{|s_j(t) - s_i(t)|}, \quad (3)$$

where p and q are real numbers. Equation (2) is

obtained by setting $p = 1$ and $q = 1$ in (3) (note that the irrelevant factors $1/2$ and G have been dropped).

A model which has given excellent empirical results is defined by (3) with $p = 0$ and $q = 0$:

$$g(i, t, dt) = dt^2 \sum_{j \in N(t), j \neq i} \frac{1}{m_i(t)} \frac{s_j(t) - s_i(t)}{|s_j(t) - s_i(t)|^3}. \quad (4)$$

This is called the *unit attraction Markovian model*. The difference between this model and the modified physical model given by (2) is that the attraction $m_i^p m_j^q$ between two particles is always 1, regardless of the masses of the particles. This has very definite intuitive appeal, in that the "similarity" between particles in the clustering sense very properly might depend only on the distance between them, and not on their size or the number of particles they represent.

This modification imparts to the model a property which is very desirable to many notions of clustering. This property is that a set of particles in a strong, closely-knit group has less tendency to "take-in" an outlying particle than does a set of less homogeneous particles, other factors being equal. The addition of such a particle to such a group might be very satisfactory for the particle but very disruptive to the group, so that a better settlement might result if the particle united with another group which it disturbed less. In the probabilistic interpretation, the particle might more likely have originated from the more scattered, less homogeneous group than from the stronger group, since it was assumed to be an outlier itself.

The importance of this property is strongly emphasized. The reason the model has the property is that a closely-knit group will converge very rapidly to one particle, which will subsequently attract outliers only as if it had a mass of one. The composite particle will in addition move only very slowly toward other particles, since its large mass will give it a large inertial drag due to the component $1/m_i(t)$. A more separated group of particles, however, will continue as multiple particles for a longer period of time, with all of the remaining particles attracting an outlier to the group. Sample clusterings using the unit attraction Markovian model are presented later in the paper, with intuitively very satisfactory results.

It seems appropriate to remark briefly about the effect of gravitational clustering on the centroid of the set of particles being clustered. It is a physical property that under true gravitational motion the centroid is a constant with respect to time (assuming the particles start at rest and there are no external forces). There is no such assurance, however, for other gravitational models. It has been shown by Wright⁽¹⁰⁾ that for the generalized Markovian model given by (3), the centroid of the set of particles will be constant if and only if $p = q$. Thus in particular the models given by (2) and (4) are centroid-preserving.

It is important to keep in mind that the preservation of the centroid is not an inherently necessary or desirable characteristic for all clustering methods. The centroid is a physical place which has meaning

and usefulness in physical models. Some clustering problems may dictate the preservation of the centroid by their nature and properties, but other problems might not necessarily imply such a restriction.

Two more examples will be mentioned briefly. In the first one, the gravitational function is given by

$$g(i, t, dt) = dt^2 \sum_{j \in N(t), j \neq i} \frac{\sqrt{m_i(t)m_j(t)}}{m_i(t)} \frac{s_j(t) - s_i(t)}{|s_j(t) - s_i(t)|^3}. \quad (5)$$

This is clearly a centroid-preserving Markovian model with $p = q = 1/2$. It is immediately seen that this is a "compromise" between the original Markovian model ($p = q = 1$) and the unit attraction Markovian model ($p = q = 0$), in that the attraction of a group of joined particles on another group of joined particles is less than the product of their masses but greater than unity (except in the special case where both masses are unity). This has the effect of reducing the tendency for a closely-knit group to reject an outlying particle, which is such a significant characteristic of the unit attraction model.

An intriguing aspect of this model is that the effect on the model due to mass is dimensionless. Thus absolute time, not just relative time, is constant with respect to magnification of the mass scale. This can be seen from the term

$$\frac{\sqrt{m_i(t)m_j(t)}}{m_i(t)},$$

in (5). Again, there is no obvious inherent necessity for such a trait in all clustering models, but it is an interesting feature.

As a second example, consider the model given by

$$g(i, t, dt) = dt^2 \sum_{j \in N(t), j \neq i} \frac{1}{\sqrt{m_i(t)}} \frac{s_j(t) - s_i(t)}{|s_j(t) - s_i(t)|^3}. \quad (6)$$

This is the same as the unit attraction Markovian model except that the "inertial drag" $I_i(t)$ is given by

$$I_i(t) = \frac{1}{\sqrt{m_i(t)}}.$$

This, like the previous model, also "compromises" the tendency for closely-knit particles to reject outliers. But whereas the previous model gained this effect by damping the attraction component and not altering the inertial component (as compared to the original Markovian model (2)), this new model reduces the attraction component all the way to unity but then dampens the inertial drag. Thus, while big clusters do not have a greater attraction due to their size, neither do they move more slowly in direct proportion to their size. It is immediately seen from (6) that this model is not centroid-preserving. It is interesting to observe, however, that if the term "centroid" is re-defined to be

$$\frac{\sum_{i=1}^n \sqrt{m_i} s_i}{\sum_{i=1}^n \sqrt{m_i}},$$

TIME = 59.674 :	POINT	MASS				
	1.	7205.	3.1432	3.3170	1.2525	3.2434
	2.	7651.	3.1786	3.1735	0.8263	3.1515
	3.	8396.	3.1543	2.8753	0.6841	2.8439
	11.	36.	4.7234	4.0566	-2.0942	8.0376
	14.	1065.	3.1327	2.2334	0.0939	2.2033
	19.	28.	4.8051	5.3371	-3.7479	11.7796
	35.	19.	2.4791	0.7943	0.0906	0.8176
	41.	82.	4.3359	3.1961	-0.3525	4.8348
	42.	12.	4.9248	4.4138	-2.5971	9.0994
	58.	18.	4.4675	3.7374	-0.7535	6.5572
	106.	4.	4.6163	3.6180	-0.2360	6.8507
	176.	4.	4.3681	4.6649	-2.2616	3.1404
	180.	1.	4.6147	6.3135	-4.2968	31.3351
TIME = 60.973 :	POINT 106.	JOINS POINT 58.	AT LOCATION			
				4.4927	3.7149	-0.6637
						6.6034
TIME = 71.589 :	POINT 42.	JOINS POINT 11.	AT LOCATION			
				4.7590	4.1350	-2.1791
						8.2436

Fig. 1.

then this model is "centroid-preserving". Thus, like true centroid-preserving models, the location of the final composite particle can be determined without executing the model.

INTERPRETATION OF RESULTS

Now that the technique of gravitational clustering has been described, its specific application to cluster analysis will be considered in greater detail. What is the optimal clustering? How many clusters does it contain? How strong is it? What are the subclusters and how strong are they? What is the complete hierarchy of subclusters? Gravitational clustering provides answers to these questions through the analysis of the movement of the elements as they "gravitate" toward each other. Several techniques have been designed for observing the process, and these will now be described briefly. For a more complete discussion see Wright.⁽¹⁰⁾

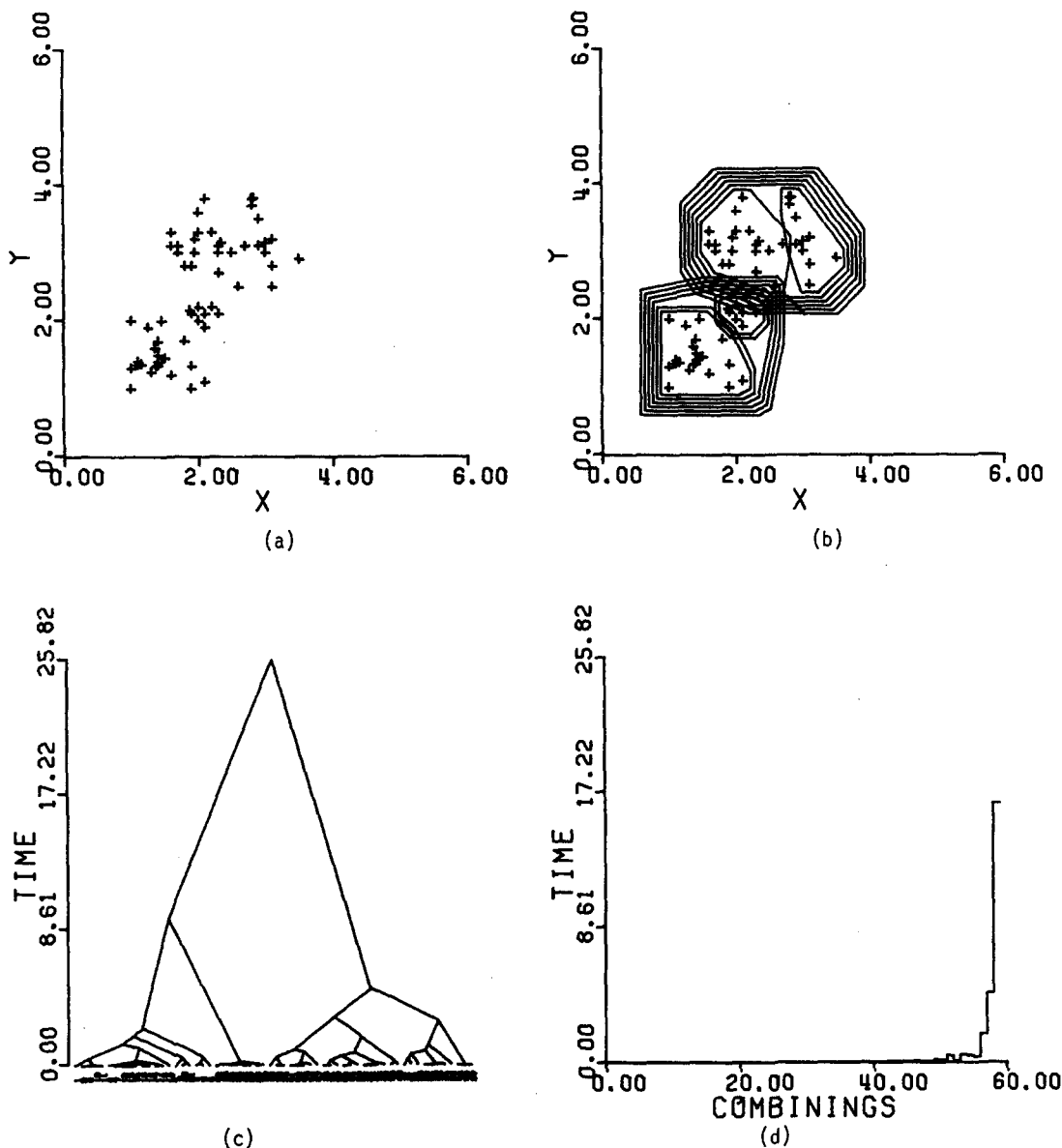
The most basic form for displaying the gravitational process is to list the occurrence of events (cf. Fig. 1). These events are of two types, the movement of a particle and the joining of two or more particles.

The contour plot (cf. Fig. 2b) is probably the most graphical of the displays presented here. Its purpose is to illustrate the clustering structure as it exists at various times in the simulated process. The points are initially plotted, and a segment of time is chosen so as to divide the total life of the system into a specified number of equal subintervals. At the end of the first subinterval the clusters which have formed up to that time are determined, and a convex polyhedron is drawn around each such cluster. This step is then repeated for the second subinterval, and so on up to the last subinterval. In this manner much

of the clustering hierarchy will be vividly portrayed, and the strengths of the clusters will also be indicated, since a relatively strong cluster might exist over several subintervals of time. An illustration of this plot has been given by Wright.⁽¹²⁾ Napior⁽⁴⁾ presents a contour-type display in which subsets are manually encircled to indicate clustering structure. These circles, however, do not directly correspond to any measured value such as time or distance, and only a single set of circles, rather than a hierarchy, is used. Rohlf⁽⁵⁾ uses manually drawn "contour circles" to indicate a hierarchy of clusters.

Clearly, the contour plot is restricted to 2-dimensional data, although some insight into 3- or 4-dimensional data may be gained by the use of projections. While such a restriction is very significant for practical clustering problems, it does not greatly diminish the value of this display for studying or testing clustering procedures, since artificial 2-dimensional data can usually be used for that purpose with no loss in generality.

The tree plot (cf. Fig. 2c) is analogous to the dendrograms described by Sokal and Sneath⁽⁷⁾ and more recently by Sneath and Sokal,⁽⁶⁾ which are a common display tool in cluster analysis. It is unrestricted as to dimensionality and yields a continuous interpretation of the state of the model. The plot consists of a binary tree in which each node corresponds to a cluster in the clustering hierarchy, and each branch indicates a hierarchical relationship. The root naturally corresponds to the entire data set, and the terminal nodes correspond to individual data elements. The cluster corresponding to a node consists of the terminal nodes in the subtree having that node as its root, and the vertical component of the node indicates the time of formation of the cluster.



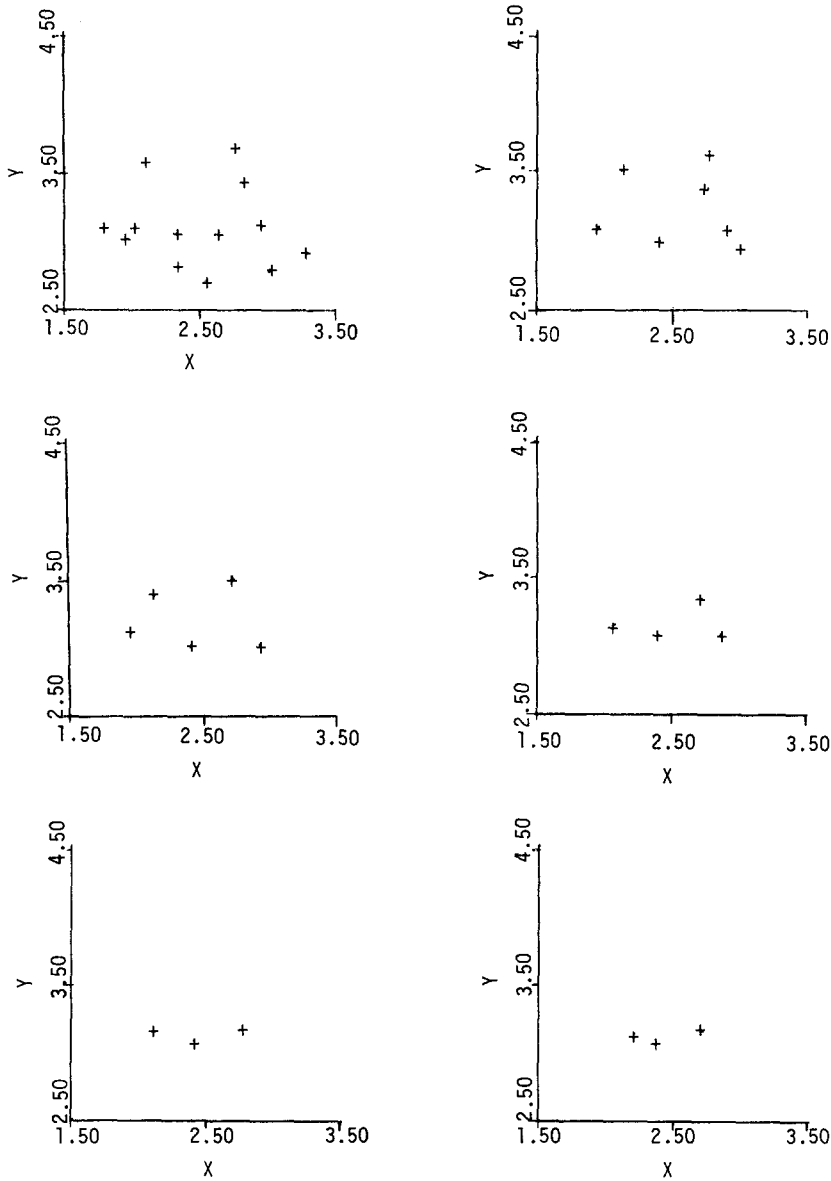
Graphical Displays for Arbitrarily
Generated Points

Fig. 2.

A time plot (cf. Fig. 2d) is a display which is used to give a crude but easily understandable picture of the relative strengths of the clustering levels. In this plot, a horizontal line segment is drawn corresponding to each binary joining. All the segments are equal in length but their height varies according to the amount of time between successive joinings. The first (left-most) line segment drawn indicates the time until the first binary joining, the second segment indicates the time between the first and second joinings, and so on until the $(n - 1)$ th segment indicates the time between the $(n - 2)$ th and the $(n - 1)$ th binary joining.

Two other displays might be used to indicate the

motion of the elements in the simulation process. One such display consists of plotting the positions of the remaining elements at regular intervals of time, and is called a position plot (cf. Fig. 3). Extension of this to the continuous case would yield a motion plot which would consist of a continual display of the moving particles on a display tube or in a movie picture. This display might be especially helpful in observing the effects of scaling adjustments on the data, comparing the motions obtained using different gravitational functions, and picking out "borderline" elements which are close to moving into more than one cluster. Note of course that these displays are also restricted to 2-dimensional data.



Position Plot

Fig. 3.

SAMPLE STUDIES

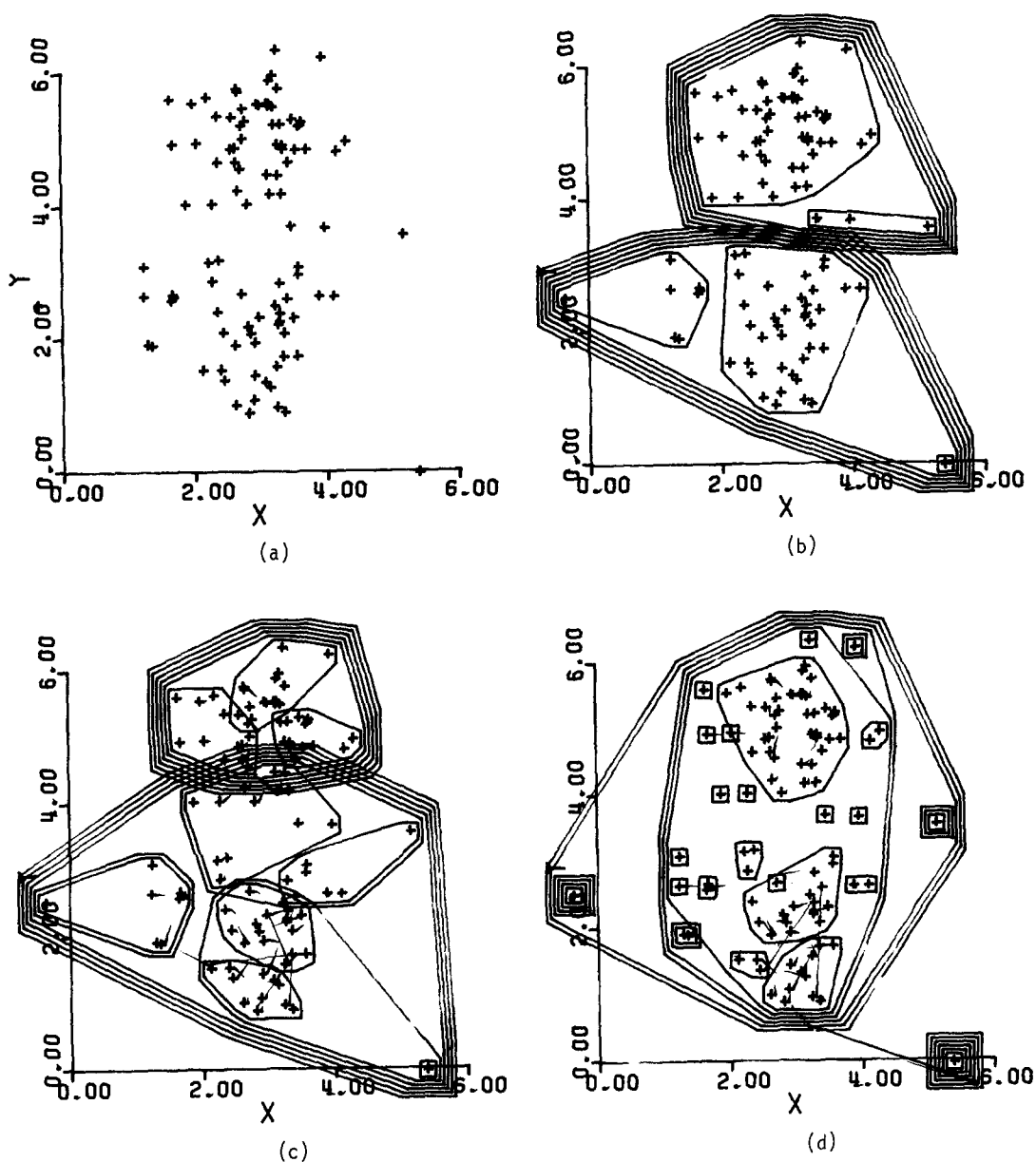
Introduction

Several data sets have been clustered using gravitational models, and a few will be presented here. The model used is the unit attraction Markovian model given by (4). For purposes of comparison, two other clustering methods which have been commonly cited in the literature will be illustrated.

The first one is called the centroid replacement method. It involves starting with n points, searching for the two which are the closest together, and combining them into a single point at their centroid. Next the new set of $n - 1$ points is considered, and the two closest points are replaced by a single point at their

centroid. The process is continued in this manner until there is only one point remaining. The method is similar to the gravitational method in that the number of points is decreasing and the sizes of the points are increasing. Another version of the method involves searching, not for the two points which minimize distance between them, but for the two points which minimize the product of the distance between times the masses of the points. This is equivalent to finding the two points with minimum sample variance. It is a major improvement over the original version, and is the version illustrated here.

The other method is called the single-link method. It involves taking a threshold distance and linking together points which lie within that threshold dis-



Two Normal Distributions

Fig. 4.

tance of each other. The clusters consist of the maximal sets of connected points, in which each point is connected to every other point by a chain of single links. The threshold is started at zero and increased until finally all the points are linked into one cluster. So again we have a sequence of events in which points or groups of points combine at certain values of a parameter (threshold) and form larger groups which are fewer in number.

All clustering programs are written in FORTRAN IV and have been run on an IBM 370/158 computer. The amount of CPU time required to execute the programs naturally depends on the size of the data

set. For the gravitational model, it also depends on the separation of the data and the maximum step size δ . Specifically, it varies directly with the number of dimensions, and inversely with δ . For most of the data sets used here, involving 100 points in 2-dimensions, the run times for the centroid replacement method were approximately 11 sec, and for the single-link method 9 sec. For the gravitational method with $\delta = 0.03$ the run times were approx. 44 sec. For the 60 point data set with $\delta = 0.03$, the gravitational run time was approx. 8 sec. The smaller time was due not so much to the smaller number of points but to the smaller separation between the points. Using

different values for δ , gravitational clustering has been done in a few minutes on data sets of 114 points in 44 dimensions and 540 points in 4 dimensions.

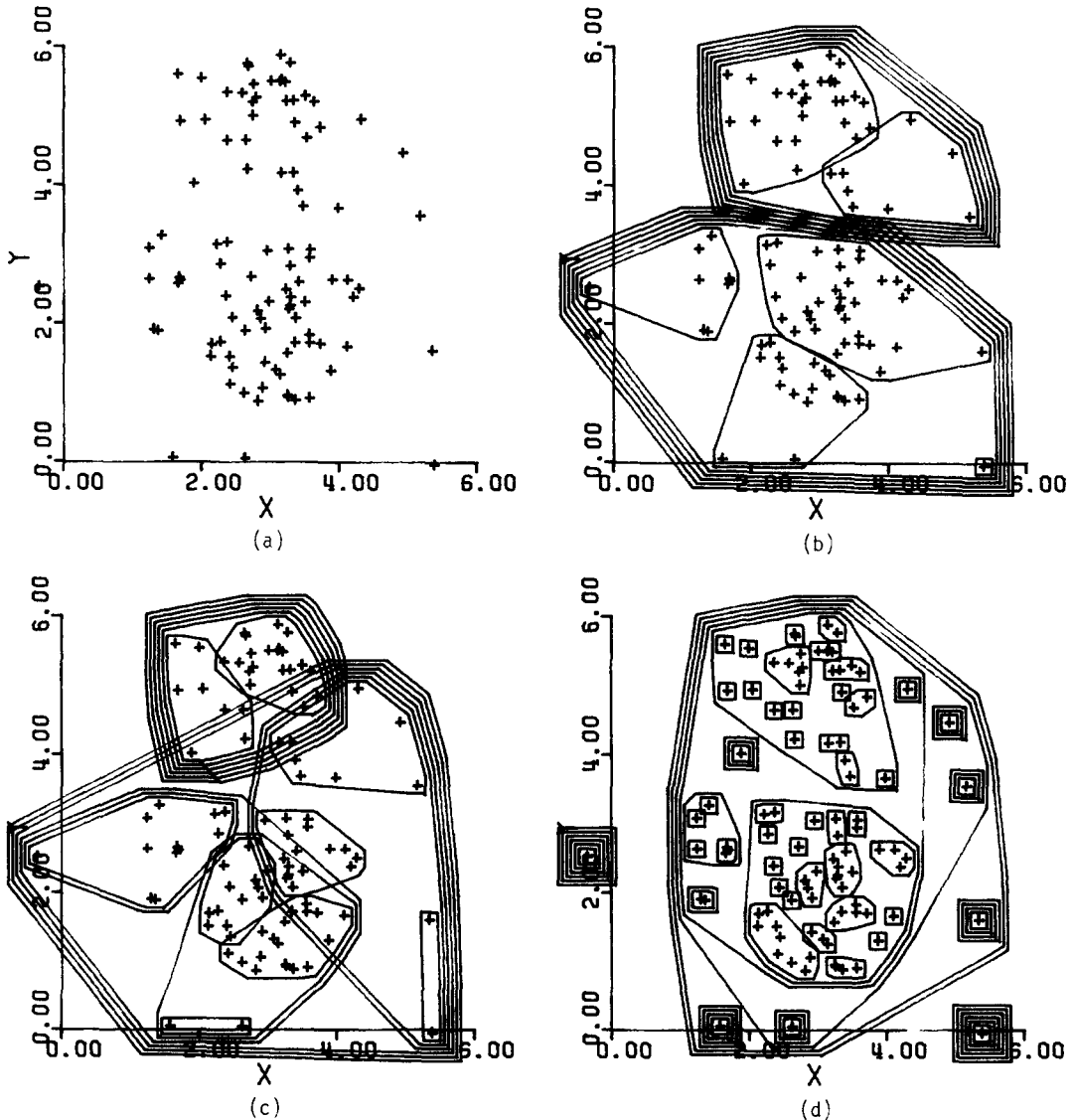
Arbitrarily generated points

The first sample to be considered consists of a set of 60 points manually marked in 2-dimensional space. The points are plotted in Fig. 2(a). The gravitational clustering produced the contour plot of Fig. 2(b) referred to earlier. The total simulated time was 62.6 sec, and the contour lines correspond to 10-sec intervals. It is clear from the contour plot that the points cluster best into two clusters, one containing 29 points and the other containing 31 points. Approximately 64% of the life of the system was spent with 2 particles remaining. The lower cluster also has two reasonably good subclusters, one containing 21 points and the other containing 8 points. These sub-

clusters were in existence for *ca.* 27 and 35%, respectively, of the life of the system.

The time plot is given in Fig. 2(d). It also indicates that the strongest clustering level is 2, the next strongest 3 (21, 8, and 31 points), and the next strongest 4 (21, 8, 20, and 11 points). After that, little significant strength is shown.

The tree plot is given in Fig. 2(c), and it provides a very complete picture of the simulation. It indicates that the system existed with 2 clusters for *ca.* 40 sec, with 3 clusters for *ca.* 11 sec, with 4 clusters for 4.5 sec etc. (This may be checked with the time plot.) The lower cluster, containing 29 points, existed for *ca.* 17 sec. Its 8-point subcluster was unusually strong, forming at time 0.7 sec and not joining with the other subcluster until time 22.5 sec, for a total life of *ca.* 22 sec. The 31-point cluster was not so well separated, dividing into subclusters at a rather uniform rate.



Different Sized Normal Distributions

Fig. 5.

Two normal distributions

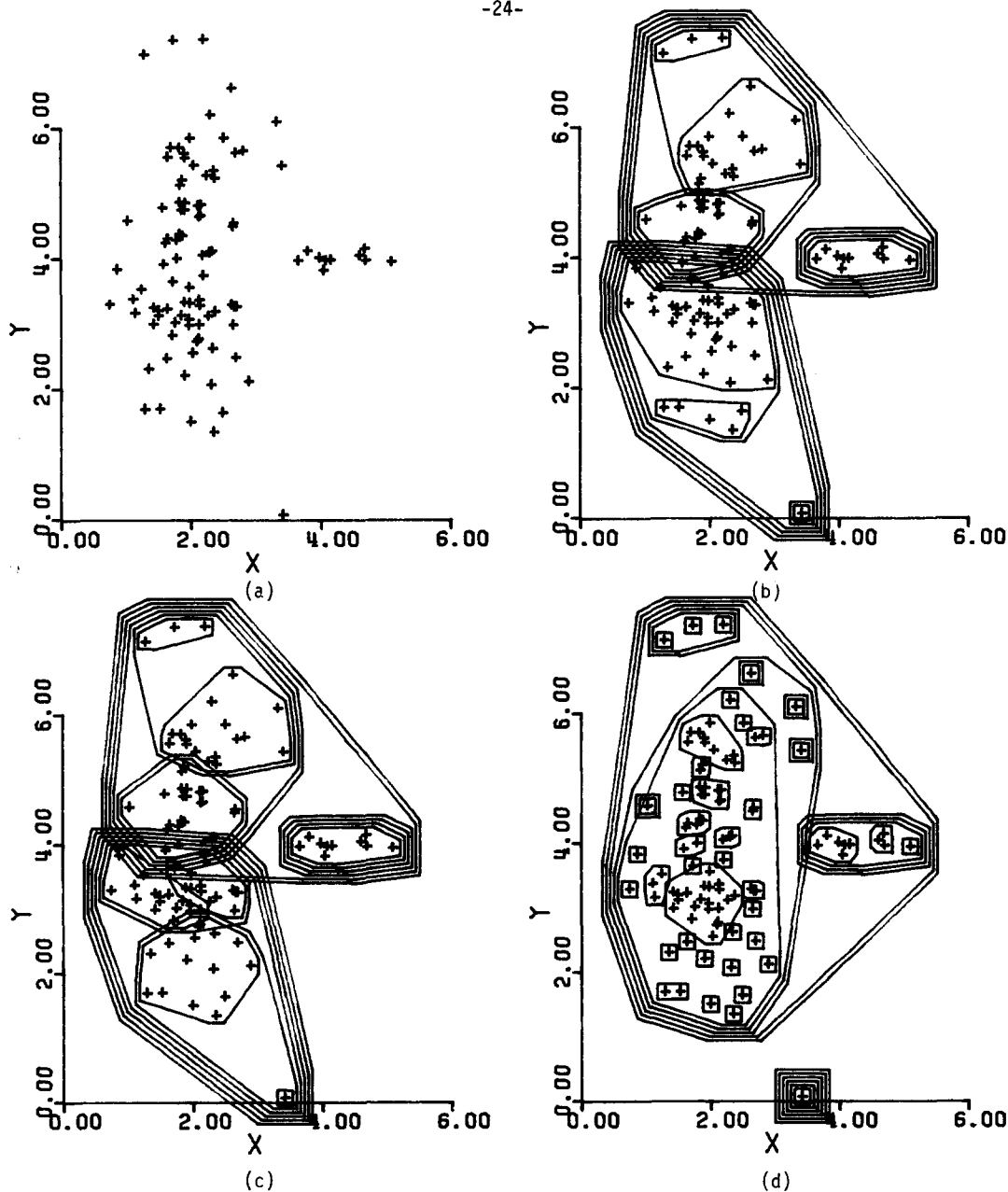
The second data set consists of 50 points each from two normal distributions with different means and different dispersions (Fig. 4a). Contour plots using the gravitational method, centroid-replacement method, and single-link method are given in Figs. 4(b), (c), and (d), respectively. Because of the closeness of the clusters, some of the contour lines overlap and make it difficult to determine the exact allocation of some of the points. Fewer contour lines might make an interpretation easier, or a tree plot could be studied to determine the exact allocation.

The gravitational method seems to pick off the separation very nicely. In the centroid replacement method, the lower cluster appears to reach too high for some of its points. The single-link method gives little indication of the clustering structure.

Different-sized populations

The third data set consists of 30 points from one normal distribution with a small dispersion and 70 points from another normal distribution with a larger dispersion (Fig. 5a). The gravitational, centroid replacement, and single link clusterings are indicated

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Two Different-Sized Elliptical Distributions

Fig. 6.

in Figs. 5(b), (c), and (d), respectively. Again, the gravitational method seems to this observer to pick up the clusters very satisfactorily, with the centroid replacement method doing some surprising allocation, and the single link method being of little help.

Different-sized elliptical distributions

The fourth data set consists of one elliptical distribution containing 90 points with a large dispersion, and another elliptical distribution containing 10 points with a small dispersion (Fig. 6a). The clusterings are given as usual in Fig. 6(b), (c) and (d). Note that both the gravitational method and the centroid replacement method pick up the small cluster, although it is also grouped at the 2-level clustering with half of the large distribution, because it is not

significant enough to comprise an independent cluster at the 2 level.

Iris species

The fifth example is quite well known and consists of measurements taken on 150 Iris flowers (c.f. Fisher⁽³⁾ and Anderson.⁽¹⁾) According to botanical classification, the first 50 samples were from the species *Iris setosa*, the next 50 from the species *Iris versicolor*, and the next 50 from the species *Iris virginica*. Previous results have indicated that the *Iris versicolor* and *Iris virginica* are closely related to each other, but are rather distinct from the *Iris setosa*. Four measurements were taken on each flower, consisting of the sepal length, the sepal width, the petal length, and the petal width.

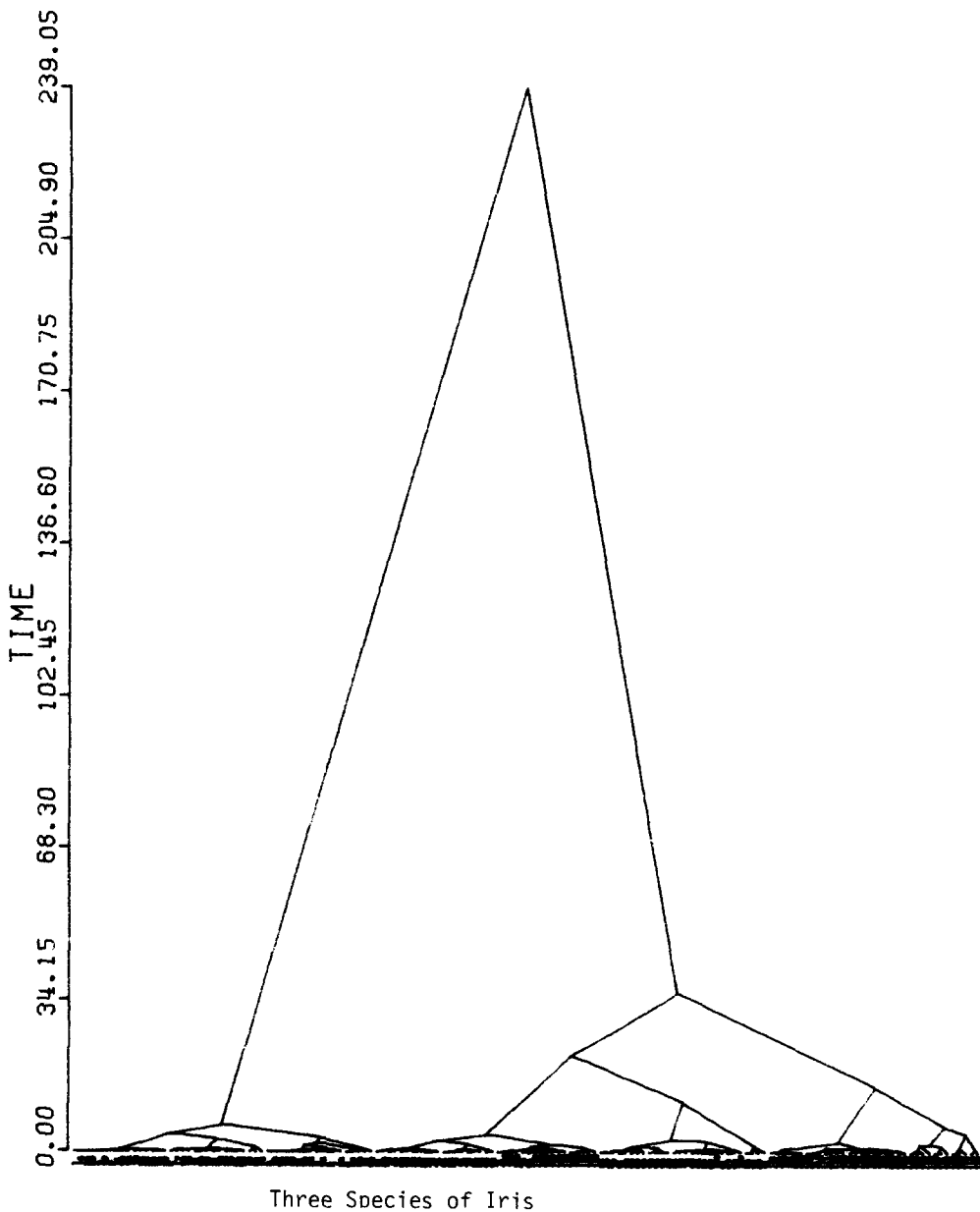


Fig. 7.

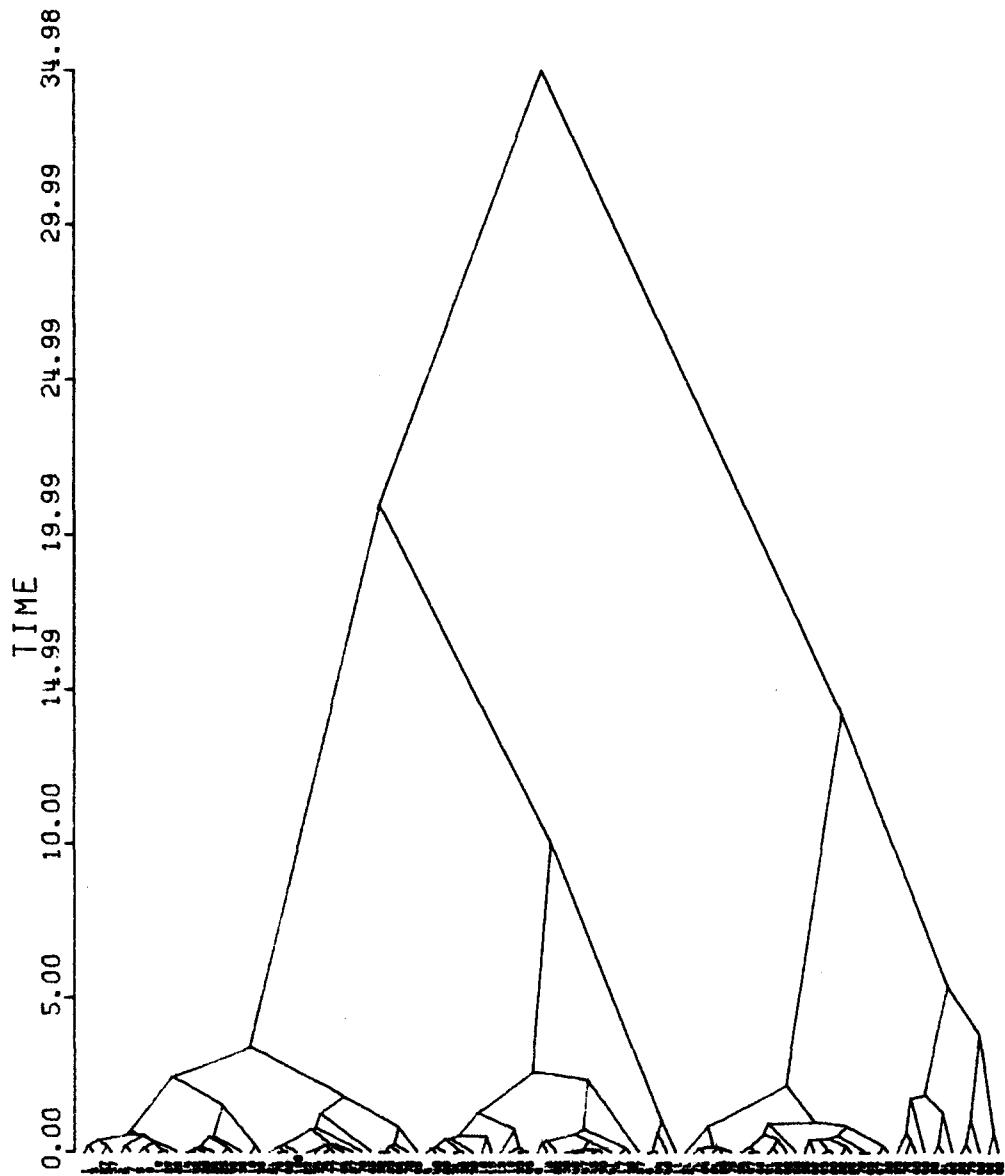
The tree plot for this sample is given in Fig. 7. The element numbers at the bottom of the plot are unreadable on this small a scale, but the essence of the results can still be understood with a little explanation. The results are consistent with the earlier indications that the *Iris setosa* is quite distinct from the other two species, since the first 50 elements come together very rapidly and then remain as an individual unit until the end. The 2-level clustering has a very high life time of $338-50/338 = 85\%$.

Because of the relative dissimilarity between the *Iris setosa* and the other two species, the model was run again only for the *Iris versicolor* and the *Iris virginica*, now numbered 1-100. The tree plot for this sample

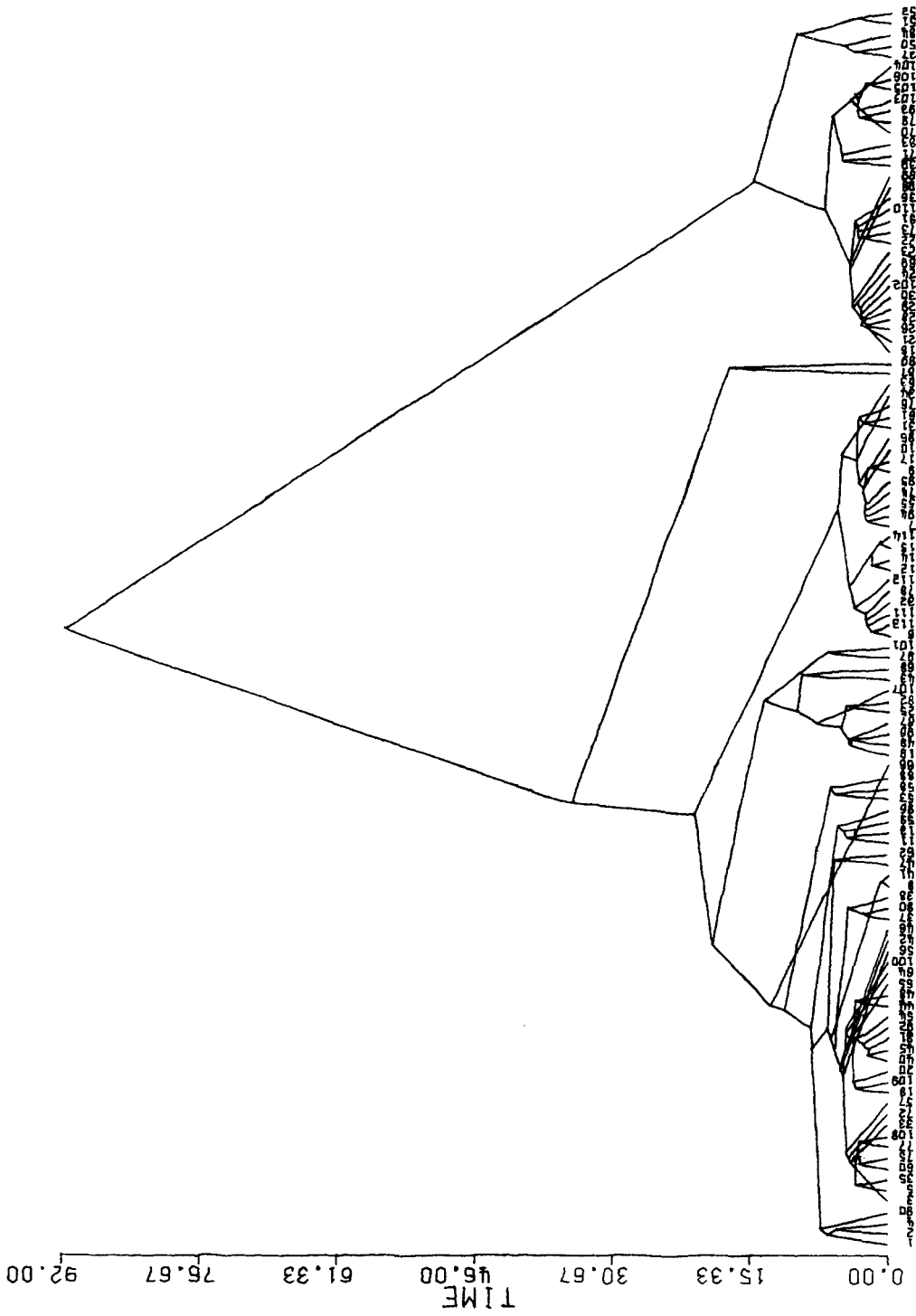
is given in Fig. 8. The life times for the 2-, 3-, 4-, and 5-level clusterings are 40, 19, 7, 12, and 25% respectively. The unusual strength at the 5 level is particularly noteworthy, and might indicate an alternative species classification. The intermixing between the two species can be seen from Fig. 8.

Occupational groupings

The sixth and final example consists of 44 traits specified for 114 occupational groupings, as obtained from (8). The occupations are listed in Table 1 and the traits are given in Table 2. The tree plot is given in Fig. 9 and is self-explanatory as to the clusterings obtained.



Two Species of Iris
Fig. 8.



Tree Plot of Occupational Groupings

Fig. 9.

Table 1. Occupational groupings

1. Arts instructive	39. Manipulating	77. Piloting/driving
2. Art-decorating	40. Nurse-supervisor	78. Legal work
3. Photography	41. Industrial tng	79. Guard/protect
4. Creative art	42. Vocational educ	80. Machine trades
5. Art-restoration	43. Flight training	81. Machine-set up
6. Bus admin	44. H.S./college teach	82. Machine-control
7. Bus negotiating	45. Elementary educ	83. Equipment driver
8. Business training	46. Instruction-misc	84. Machine tending
9. Bus supervisory	47. Physical educ	85. Supervise service
10. Bus management	48. Supervisory tng	86. Health physics
11. Bus consulting	49. Animal train	87. Scientist
12. Interviewing	50. Signaling	88. Math/science
13. Accounting	51. Machine feeding	89. Surgery
14. Legal documenting	52. Handling	90. Medical
15. Corresponding	53. Engineer/design	91. Therapy
16. Infor management	54. Sales-engineer	92. Nurse/med tech
17. Schedule/dispatch	55. Production manage	93. Child-adult care
18. Secretarial	56. Drafting	94. Public relations
19. Allocate/expedite	57. Technical work	95. Purchasing/sales
20. Paying/receiving	58. Engineer	96. Sales/service
21. Cashiering	59. Ind engineer	97. Demonst/sales
22. Inspecting	60. Surveying	98. Delivery/service
23. Bus mach operator	61. Technical writing	99. Sales-specialized
24. Classify/filing	62. Entertainer	100. Creative music
25. Stenography	63. Dramatics	101. Beautician/barber
26. Computing clerk	64. Music-instrument	102. Customer service
27. Clerical sorting	65. Musical-vocal	103. Service/misc
28. Clerical typing	66. Music-rhythmics	104. Accommodating
29. Clerical checking	67. Announcing rad/TV	105. Personal service
30. Switchboard	68. Performing	106. Messenger/usher
31. Behavior science	69. Amusement	107. Animal care
32. Counseling	70. Special entertain	108. Photomach work
33. Crafts-foreman	71. Modeling	109. Techwork-radio/TV
34. Crafts-supervisor	72. Farm-plant/animal	110. Public transport
35. Tailoring	73. Farming-technical	111. Journalism
36. Cooking	74. Inspect/protect	112. Creative writing
37. Craftsmanship	75. Laboratory work	113. News reporting
38. Precision work	76. Appraise/inspect	114. Translate/edit

CONCLUDING REMARKS

These six examples give only a glimpse into the performance of the unit attraction Markovian gravitational clustering model. Many more examples have been run, and the results have been completely reasonable in the intuitive judgement of the author. Obviously, intuition is a very imprecise basis for making judgements, especially when there is a potential for bias created by judging one's own method. However, it can contribute to significant evaluation of clustering procedures, especially if there is a consensus of opinion. It is hoped that this presentation will afford a general though limited opportunity for such evaluation.

An empirical comparison of the gravitational model with the two other clustering methods has been very useful in the evaluation of all three. The examples presented here are typical of all examples run, and were selected to illustrate results using different types of data. The gravitational method and the centroid replacement method seemed to give quite reasonable results, but the results of the single-link method were virtually useless for the sort of clustering considered here. Certainly there are other linkage methods which should be considered, and the results

using this method would not necessarily be similar to results using other linkage methods.

Concerning a comparison between the gravitational and centroid replacement methods, it seems to this observer that for several data sets the gravitational method gives better clusterings especially for borderline points. Naturally the centroid replacement method is faster, but if it is felt that the gravitational method gives the best results, and if the overriding concern is to obtain the best clustering, then the extra expenditure of time to run the gravitational model may be of little concern.

It should be pointed out, moreover, that the speed of the gravitational method can be increased by increasing δ , and that other values of δ could have been used in these examples, with different timing results. As δ is increased, of course, the motion computed by the model will be a poorer approximation of the theoretical motion. To some extent, it is possible to look upon the gravitational model as a generalization, to a continuous process, of the centroid replacement method, in which two points are replaced by a single point at their centroid only after continuous motion has occurred and they have gotten very close together.

Table 2. Occupational traits

Number	Meaning
<i>Education</i>	
1	General educational development
2	Specific vocational preparation
<i>Aptitude</i>	
3	Intelligence
4	Verbal
5	Numerical
6	Spatial
7	Form perception
8	Clerical perception
9	Motor coordination
10	Finger dexterity
11	Manual dexterity
12	Eye-hand-foot coordination
13	Color perception
<i>Temperaments</i>	
14	Variety and change
15	Repetition and short cycle
16	Under specific instruction
17	Direction, control, and planning
18	Dealing with people
19	Influencing people
20	Performing under stress
21	Sensory or judgmental criteria
22	Measurable or verifiable criteria
23	Interpretation of feelings, ideas, or facts
24	Set limits, tolerances, or standards
<i>Interests</i>	
25	Things and objects
26	Business contact with people
27	Routine concrete, organized activities
28	Social welfare or dealing with people and language in social situations
29	Prestige or esteem of others
30	People and the communication of ideas
31	Scientific or technical activities
32	Abstract or creative activities
33	Nonsocial activities-processes, machines, techniques
34	Tangible, productive satisfactions
<i>Physical capabilities</i>	
35	Sedentary work
36	Light work
37	Medium work
38	Heavy work
39	Very heavy work
40	Climbing, balancing
41	Stooping, kneeling, crouching, crawling
42	Reaching, handling, fingering, feeling
43	Talking, hearing
44	Seeing

Many further illustrations could have been given, some extending the path of earlier models, others completely divergent from earlier models. For example, there was no specific mention of a model with a distance component being inversely proportional to the absolute Euclidean distance, instead of to the Euclidean distance squared. Nor was there any discussion about a model having an attractive force between two masses being proportional to the *sum* of the two masses, i.e. to the total mass involved.

In general, a person who does gravitational clustering must necessarily choose the specific model he wishes to use. Sometimes the particular characteristics of his information will dictate some of the features

of the model. Usually, however, he will merely select features that intuitively seem to be desirable in general, such as the Markovian property, the unit attraction property, and the centroid-preservation property.

A significant amount of theoretical evaluation of the gravitational clustering method has been carried out, although it will only be alluded to here. Wright^(11,13) has proposed a formal model for cluster analysis that consists to a large extent of specifying several properties that all legitimate clustering methods should satisfy. Because of the heuristic implementation of the gravitational model, it is impossible to determine theoretically whether or not one of the properties, regarding continuity, is satisfied. In addition, a

straightforward extension of the model, regarding its "domain of definition," must be made in order to satisfy another property. With these exceptions noted, however, it has been shown theoretically that the unit attraction Markovian model satisfies all other properties.⁽¹⁰⁾

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